

CV

TL;DR

I'm Luke - a software engineer and engineering lead determined to solve every problem collaboratively and elegantly. I firmly believe in the power of simplicity and curiosity.

My primary stack consists of: TypeScript, Node.js, Bun, AWS and Claude Code.

You can checkout my portfolio at lukehedger.dev and github.com/lukehedger

Experience

Over the past 19 years, I have worked at all layers of the software engineering stack; on serverless payment platforms, state-of-the-art React apps, blockchain networks, viral marketing websites, open source tools and a pineapple delivery system. I've worked for early-stage startups, international corporations, prominent charities, infamous advertising agencies and one of the biggest brands on the planet. I've also written several articles and given many talks about the things I've discovered along the way.

In 2024, I co-authored the book *Serverless Development on AWS*, published by O'Reilly.

I was recognised as an AWS Community Builder from 2022-2026 for contributions to the serverless community.

Timeline

Jul 2021 - Current | **Lead Software Engineer (Payments and Fraud Prevention)** | The LEGO Group

Lead engineering on the payments platform that processes transactions across the LEGO Group's digital business units, including LEGO.com, the LEGO Insiders app and LEGO Education. Technical lead for a team of 7 engineers, setting direction on architecture, reliability and delivery.

The platform is fully serverless on AWS, built around an event-sourced payments lifecycle engine. The stack leverages Lambda, Lambda Durable Functions, Step Functions, EventBridge, SQS, DynamoDB, S3 and X-Ray, with a third-party fraud prevention vendor integrated into the transaction flow.

Over 5 consecutive Black Friday / Cyber Monday peak weekends, the platform has scaled and processed sales events without incident - part of a wider track record of highly successful launches across the business. As one of the early adopters of serverless architecture at scale, the codebase has matured alongside community best practice, continuously evolving as the ecosystem has grown.

Beyond the platform, I've grown several internal communities - a backend engineering meetup, a whitepaper discussion group and a lead engineering community - and mentored senior engineers across the organisation. I've authored guidelines on code quality, test strategy, reliability, agentic AI usage, CDK, TypeScript and serverless development that are used by engineering teams across the group.

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Sep 2019 - June 2021 | **Lead Software Engineer (Payments)** | Cancer Research UK

Lead development of a greenfield payments platform to process Cancer Research UK's online donations, including the Stand Up To Cancer TV campaigns on Channel 4. Provided technical leadership on architecture, implementation and operation in response to critical demand from the charity to migrate processing of 1000s of daily transactions from an unreliable and expensive legacy system to a resilient, scalable and cost-effective platform.

In the first few months, the platform realised a 94% reduction in operational costs, scaling without incident to process over £1m of donations with a 30% increase in conversion.

The payments platform is cloud native on AWS and fully serverless, with an event-driven architecture that leverages Lambda (Node.js/TypeScript), Step Functions, Amplify, AppSync, API Gateway, EventBridge, SQS, Kinesis Firehose, Glue and Athena.

A rapid yet sustainable agile workflow was established to continually deliver new features and optimisations, combining the modern practices of infrastructure-as-code, continuous delivery, GitOps and fully automated, self-mutating deployment pipelines with tools such as the AWS CDK, AWS CodePipeline and GitHub Actions.

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Mar 2017 - Aug 2019 | **Software Engineer, Head of Platform** | JAAK

Headed up development of the JAAK platform, providing content infrastructure to innovative music platforms and independent artists, with features including user authentication, content delivery and payments. Worked across the entire stack to build scalable APIs in Go and performant React applications that aggregated rights data and assets for millions of musical works and recordings. Also lead research projects around programmatic licensing on the Ethereum blockchain, modelling decentralised identity and reputation in an open network and developing a rules engine for verifiable claims using JSON Schema and Lua.

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Apr 2016 - Mar 2017 | **Software Engineer** | Unlease

Lead product development of web application for UK-wide peer-to-peer rental community. Consistently shipped regular feature updates to 1000s of users, such as KYC, live text and video chat and intuitive date-picker. I also built a bunch of internal tools, including an account management dashboard, automated sitemap generator and pineapple delivery logistics system for a promotional campaign where Londoners could send each other pineapples.

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Oct 2014 - Apr 2016 | **Frontend Developer** | M&C Saatchi

Worked in agile, multi-discipline teams to deliver smart, modern websites and applications for clients including Converse, Land Rover, Ikea and Compare the Market.

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May 2013 - Oct 2014 | **Creative Developer** | Lean Mean Fighting Machine

Worked on briefs from ideation through to deployment, churning out finely crafted creations for clients like Unilever, Men's Health Forum, Forevermark and Google.

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Nov 2010 - May 2013 | **Lead Systems Engineer** | Enrich Reward Ltd.

Lead the conception, design and development of an enterprise-grade employee benefits SaaS platform that serviced 1000s of users.

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Nov 2007 - Nov 2010 | **Web Analyst** | Mercer Ltd.

Reporting and maintenance of global pensions administration system.